

Assignment: Linear Regression & Logistic Regression (Using Real Datasets)

Objective

This assignment helps students understand **Linear Regression** and **Logistic Regression** by applying them to **different real-world datasets**. Students will practice data preprocessing, visualization, model training, evaluation, and result interpretation using Python and scikit-learn.

Tools & Libraries

You are required to use the following libraries:

- Python 3.x
 - NumPy
 - Pandas
 - Matplotlib / Seaborn
 - scikit-learn
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Part 1: Linear Regression (Real Dataset)

Dataset

Advertising Dataset (Marketing / Business dataset)

(<https://www.statlearning.com/s/Advertising.csv>)

Dataset Description

The dataset contains advertising budgets for different media channels and corresponding sales values.

Features:

- TV: Advertising budget spent on TV (in thousands)
- Radio: Advertising budget spent on Radio
- Newspaper: Advertising budget spent on Newspaper

Target:

- Sales: Product sales (in thousands of units)

Problem Statement

Build a Linear Regression model to **predict product sales based on advertising budgets**.

Tasks

1. Load the Dataset

- Load the Advertising dataset using Pandas
- Display the first 5 rows

2. Data Exploration

- Check dataset shape
- Display column names
- Check for missing values
- Display summary statistics using `.describe()`

3. Feature Selection

- Independent variables (X): TV, Radio, Newspaper
- Target variable (y): Sales

4. Data Visualization

- Plot scatter plots between each feature and Sales
- Explain the observed relationships

5. Train-Test Split

- Split the data into 80% training and 20% testing

6. Model Training

- Train a **Multiple Linear Regression** model
- Print model coefficients and intercept

7. Prediction

- Predict sales values for the test set

8. Model Evaluation

- Calculate:
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score

9. Interpretation

- Explain how each advertising channel affects sales
 - Comment on model performance
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Part 2: Logistic Regression (Real Dataset)

Dataset

Titanic Dataset (Real-world classification dataset)

(<https://www.kaggle.com/c/titanic/data>)

Dataset Description

The dataset contains information about passengers on the Titanic.

Features (examples):

- Pclass: Passenger class
- Sex: Gender
- Age: Passenger age
- Fare: Ticket fare

Target:

- Survived: 1 = Survived, 0 = Not Survived

Problem Statement

Build a Logistic Regression model to **predict whether a passenger survived**.

Tasks

1. Load the Dataset

- Load the Titanic dataset using Pandas
- Display the first 5 rows

2. Data Cleaning

- Handle missing values (Age, Embarked, etc.)
- Encode categorical variables (Sex, Embarked)

3. Data Exploration

- Display survival rate
- Visualize survival by gender and class

4. Feature Selection

- Select appropriate features for prediction
- Target variable: Survived

5. Data Preprocessing

- Apply **StandardScaler** where needed

6. Train-Test Split

- Split the dataset into 80% training and 20% testing

7. Model Training

- Train a **Logistic Regression** model

8. Prediction

- Predict survival classes
- Predict survival probabilities

9. Model Evaluation

- Calculate:
 - Accuracy
 - Confusion Matrix

10. Interpretation

- Interpret the confusion matrix
 - Discuss which features most influence survival
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Submission Guidelines

- Submit a Jupyter Notebook (.ipynb)
- Include comments and explanations
- Ensure all plots have titles and labels

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- 1- <https://www.kaggle.com/c/titanic/data>
 - 2- <https://www.statlearning.com/s/Advertising.csv>
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End of Assignment